

human

Which buildings in Exeter already have solar panels on roofs?

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'User wants to find which buildings in Exeter already have solar panels on roofs. Provide the general steps to solve this geospatial query.'}

planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to be searched is buildings with solar panels on their roofs. The query does not specify a condition regarding the number of buildings, so we will aim to find as many entries as possible.

output steps: ["Find Exeter 1 area", "Search for buildings in Exeter with solar panels on roofs as many search results"]

host_agent

Tool send_message returned result : The area specified is Exeter, so there will be 1 entry for Exeter. The entity to be searched is buildings with solar panels on their roofs. The query does not specify a condition regarding the number of buildings, so we will aim to find as many entries as possible.

output steps: ["Find Exeter 1 area", "Search for buildings in Exeter with solar panels on roofs as many search results"]

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']}

Tool generate_metadata_for_artifacts returned result : (metadata)

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Search for polygons where any of the name columns contain 'Exeter' (case-insensitive)
    mask = (
        gdf['name1_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name2_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name3_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name4_text'].str.contains('Exeter', case=False, na=False)
    )
    filtered = gdf[mask]
    summary = f"Filtered polygons for Exeter. Number of matching polygons: {len(filtered)}."
    description = "GeoDataFrame containing only the area polygons whose name columns contain 'Exeter'."
    return [summary, 'filtered_exeter_area_polygon', description, filtered]
```

Tool code_executor returned result : ['Filtered polygons for Exeter. Number of matching polygons: 1.',]

Filtered polygons for Exeter. Number of matching polygons: 1. An artifact has been generated with the name: filtered_exeter_area_polygon and description: GeoDataFrame containing only the area polygons whose name columns contain 'Exeter'.

named_area_agent

The specific area polygon for Exeter has been filtered and saved as filtered_exeter_area_polygon.

host_agent

Tool send_message returned result : ['The specific area polygon for Exeter has been filtered and saved as filtered_exeter_area_polygon.Additionally some data artifacts have been generated with names ['filtered_exeter_area_polygon\'] and \n descriptions ["GeoDataFrame containing only the area polygons whose name columns contain 'Exeter'.']', []]

Calling tool generate_metadata_for_all_artifacts with args : {}

Tool generate_metadata_for_all_artifacts returned result : (metadata)

Calling tool send_message with args : {'target': 'buildings', 'task_description': "Search for buildings in the artifact 'filtered_exeter_area_polygon' that have solar panels on roofs."}

buildings_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'exeter_buildings_with_solar_panels'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['buildings_exeter_buildings_with_solar_panels', 'buildingline_exeter_buildings_with_solar_panels', 'buildingpart_exeter_buildings_with_solar_panels'],\n Descriptions are : ["A geopandas dataframe containing building data with filters and bbox applied for the query Search for buildings in the artifact 'filtered_exeter_area_polygon' that have solar panels on roofs. using filters None.", "A geopandas dataframe containing building line data with filters and bbox applied for the query Search for buildings in the artifact 'filtered_exeter_area_polygon' that have solar panels on roofs. using filters None.", "A geopandas dataframe containing building part data with filters and bbox applied for the query Search for buildings in the artifact 'filtered_exeter_area_polygon' that have solar panels on roofs. using filters None."],\n counts of records fetched are : [69169, 69342, 85480].\n ', [, ,]]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifacts 'buildings_exeter_buildings_with_solar_panels', 'buildingline_exeter_buildings_with_solar_panels', and 'buildingpart_exeter_buildings_with_solar_panels' to identify buildings with solar panels on roofs. Save the filtered results as 'filtered_solar_panel_buildings_exeter'. Focus on features related to solar panels on roofs."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['buildings_exeter_buildings_with_solar_panels', 'buildingline_exeter_buildings_with_solar_panels', 'buildingpart_exeter_buildings_with_solar_panels']}

Tool generate_metadata_for_artifacts returned result : (metadata)

Tool Code Executor

```
def filter_solar_panel_buildings_exeter(data: list):
    import pandas as pd
    import geopandas as gpd
    # Unpack the dataframes
```

```

buildings_gdf, buildingline_gdf, buildingpart_gdf = data

# For buildings, filter for solar panels on roofs
# The relevant column is 'roofmaterial_solarpanelpresence' and should be 'Present'
if 'roofmaterial_solarpanelpresence' in buildings_gdf.columns:
    filtered_buildings = buildings_gdf[buildings_gdf['roofmaterial_solarpanelpresence'] == 'Present']
else:
    filtered_buildings = buildings_gdf.iloc[0:0] # Empty GeoDataFrame if column not present

# For building parts, check if any part has solar panels on the roof
if 'roofmaterial_solarpanelpresence' in buildingpart_gdf.columns:
    filtered_buildingparts = buildingpart_gdf[buildingpart_gdf['roofmaterial_solarpanelpresence'] == 'Present']
else:
    filtered_buildingparts = buildingpart_gdf.iloc[0:0]

# For building lines, there is no solar panel info, so skip
# Combine filtered results
filtered_combined = pd.concat([filtered_buildings, filtered_buildingparts], ignore_index=True)
if isinstance(filtered_combined, pd.DataFrame) and not isinstance(filtered_combined, gpd.GeoDataFrame):
    # If concat lost geometry, convert back to GeoDataFrame
    filtered_combined = gpd.GeoDataFrame(filtered_combined, geometry='geometry', crs=buildings_gdf.crs)

summary = f"Filtered buildings and building parts with solar panels on roofs. Total entries: {len(filtered_combined)}."
artifact_name = 'filtered_solar_panel_buildings_exeter'
artifact_description = 'GeoDataFrame of buildings and building parts in Exeter with solar panels present on roofs.'
return [summary, artifact_name, artifact_description, filtered_combined]

```

Tool code_executor returned result : ['Filtered buildings and building parts with solar panels on roofs. Total entries: 4468.',]

Filtered buildings and building parts with solar panels on roofs. Total entries: 4468. Artifact name: filtered_solar_panel_buildings_exeter Description: GeoDataFrame of buildings and building parts in Exeter with solar panels present on roofs.

buildings_agent

I have identified buildings in the 'filtered_exeter_area_polygon' that have solar panels on roofs. The filtered results are saved in the artifact named '**filtered_solar_panel_buildings_exeter**'. This GeoDataFrame contains 4,468 entries of buildings and building parts with solar panels.

host_agent

Calling tool send_message with args : {'target': 'structures_agent', 'task_description': "Search for solar panel structures in the artifact 'filtered_exeter_area_polygon'."}

structures_agent

Calling tool call_os_ngd with args : {'filters': ['Made Sealed Surface Covered with Solar Panels', 'Made Unsealed Surface Covered with Solar Panels', 'Solar Panels', 'Solar Panels On Bare Earth Or Grass'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'solar_panel_structures_exeter'}

Tool call_os_ngd returned result : ['Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['solar_panel_structures_exeter'],\n Descriptions are : ["A geopandas dataframe containing structure data with filters and bbox applied for the query Search for solar panel structures in the artifact 'filtered_exeter_area_polygon'. using filters ['Made Sealed Surface Covered with Solar Panels', 'Made Unsealed Surface Covered with Solar Panels', 'Solar Panels', 'Solar Panels On Bare Earth Or Grass']. Further Name filtering is available for this"],\n counts of records fetched are : [195].\n ', []]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Please analyze the dataset 'solar_panel_structures_exeter' to filter and summarize the solar panel structures. Save the filtered results as 'filtered_solar_panel_structures_exeter'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['solar_panel_structures_exeter']}

Tool generate_metadata_for_artifacts returned result : (metadata)

Tool Code Executor

```
def filter_and_summarize_solar_panel_structures(data: list):
    import pandas as pd
    import geopandas as gpd

    gdf = data[0]
    # Filter for solar panel structures using multiple relevant columns
    filtered = gdf[(gdf['description'].str.contains('Solar Panel', case=False, na=False)) |
                  (gdf['oslandcovertierb'].str.contains('Solar Panel', case=False, na=False)) |
                  (gdf['oslandusetierb'].str.contains('Energy Generation', case=False, na=False)) |
                  (gdf['nlud_groupdescription'].str.contains('Energy production', case=False, na=False))]

    # Summarize
    summary = {
        'total_structures': len(filtered),
        'total_area_m2': filtered['geometry_area_m2'].sum(),
        'unique_descriptions': filtered['description'].unique().tolist(),
```

```

        'capture_methods': filtered['geometry_capturemethod'].unique().tolist(),
        'date_range': [str(filtered['geometry_evidencedate'].min()), str(filtered['geometry_evidencedate'].max())]
    }

    # Attach summary as attribute for output
    filtered.attrs['summary'] = summary

    return [
        f"Filtered {len(filtered)} solar panel structures. Total area: {summary['total_area_m2']} m2.",
        'filtered_solar_panel_structures_exeter',
        'Filtered GeoDataFrame of solar panel structures in Exeter with summary statistics attached as attributes.',
        filtered
    ]

```

Tool code_executor returned result : ['Filtered 1 solar panel structures. Total area: 10393.775 m2.',]

Filtered 1 solar panel structure in Exeter. The total area covered by these structures is 10,393.78 m². The filtered results have been saved as 'filtered_solar_panel_structures_exeter', which contains a GeoDataFrame of the relevant solar panel structures along with summary statistics.

structures_agent

I found and filtered solar panel structures in the 'filtered_exeter_area_polygon'. Here's the summary:

- **Total Structures Found:** 1
- **Total Area Covered:** 10,393.78 m²

The filtered results are saved as 'filtered_solar_panel_structures_exeter'.

host_agent

Calling tool generate_metadata_for_all_artifacts with args : {}

Tool generate_metadata_for_all_artifacts returned result : (metadata)

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot the buildings and building parts in Exeter with solar panels on roofs (artifact: 'filtered_solar_panel_buildings_exeter') and the solar panel structures in Exeter (artifact: 'filtered_solar_panel_structures_exeter') on the map of Exeter (artifact: 'filtered_exeter_area_polygon'). Provide a summary of the results."}

plotting_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['filtered_solar_panel_buildings_exeter', 'filtered_solar_panel_structures_exeter', 'filtered_exeter_area_polygon']}

Tool generate_metadata_for_artifacts returned result : (metadata)

Tool Code Executor

```
def plot_solar_panel_buildings_and_structures_exeter(data: list):  
    import geopandas as gpd  
    import folium  
    from shapely.geometry import mapping  
  
    # Unpack the data  
    gdf_buildings = data[0].copy()  
    gdf_structures = data[1].copy()  
    gdf_exeter = data[2].copy()  
  
    # Ensure all CRS are EPSG:4326 for folium  
    gdf_buildings = gdf_buildings.to_crs(epsg=4326)  
    gdf_structures = gdf_structures.to_crs(epsg=4326)  
    gdf_exeter = gdf_exeter.to_crs(epsg=4326)  
  
    # Get centroid of Exeter for map initialization  
    exeter_centroid = gdf_exeter.geometry.centroid.iloc[0]  
    m = folium.Map(location=[exeter_centroid.y, exeter_centroid.x], zoom_start=12, tiles='cartodbpositron')  
  
    # Plot Exeter boundary  
    folium.GeoJson(  
        gdf_exeter.geometry,  
        name='Exeter Boundary',  
        style_function=lambda x: {'color': 'black', 'weight': 2, 'fillOpacity': 0.05}  
    ).add_to(m)  
  
    # Plot buildings with solar panels  
    folium.GeoJson(  
        gdf_buildings.geometry,  
        name='Buildings with Solar Panels',  
        style_function=lambda x: {'color': 'orange', 'weight': 1, 'fillColor': 'orange', 'fillOpacity': 0.5},  
        tooltip=folium.GeoJsonTooltip(fields=['description', 'roofmaterial_solarpanelpresence'])  
    ).add_to(m)  
  
    # Plot solar panel structures  
    folium.GeoJson(  
        gdf_structures.geometry,
```

```

        name='Solar Panel Structures',
        style_function=lambda x: {'color': 'blue', 'weight': 2, 'fillColor': 'blue', 'fillOpacity': 0.3},
        tooltip=folium.GeoJsonTooltip(fields=['description'])
    ).add_to(m)

    folium.LayerControl().add_to(m)

# Save map
map_filename = 'exeter_solar_panel_buildings_and_structures.html'
m.save(map_filename)

# Prepare summary
summary = f"There are {len(gdf_buildings)} buildings/building parts with solar panels on roofs and {len(gdf_structures)} solar panel structures in Exeter.\n"
summary += "First 5 buildings:\n" + str(gdf_buildings[['description', 'roofmaterial_solarpanelpresence']].head(5)) + "\n"
summary += "First 5 solar panel structures:\n" + str(gdf_structures[['description']].head(5))

return [summary, map_filename, "Folium map showing Exeter boundary, buildings/building parts with solar panels, and solar panel structures.", map_filename]

```

Tool code_executor returned result : Traceback (most recent call last): ... AssertionError: The field description is not available in the data. Choose from: ().

Tool Code Executor

```

def plot_solar_panel_buildings_and_structures_exeter(data: list):
    import geopandas as gpd
    import folium

    # Unpack the data
    gdf_buildings = data[0].copy()
    gdf_structures = data[1].copy()
    gdf_exeter = data[2].copy()

    # Ensure all CRS are EPSG:4326 for folium
    gdf_buildings = gdf_buildings.to_crs(epsg=4326)
    gdf_structures = gdf_structures.to_crs(epsg=4326)
    gdf_exeter = gdf_exeter.to_crs(epsg=4326)

    # Get centroid of Exeter for map initialization
    exeter_centroid = gdf_exeter.geometry.centroid.iloc[0]
    m = folium.Map(location=[exeter_centroid.y, exeter_centroid.x], zoom_start=12, tiles='cartodbpositron')

    # Plot Exeter boundary
    folium.GeoJson(
        gdf_exeter.geometry,
        name='Exeter Boundary',
        style_function=lambda x: {'color': 'black', 'weight': 2, 'fillOpacity': 0.05}
    ).add_to(m)

```

```
# Plot buildings with solar panels
folium.GeoJson(
    gdf_buildings.geometry,
    name='Buildings with Solar Panels',
    style_function=lambda x: {'color': 'orange', 'weight': 1, 'fillColor': 'orange', 'fillOpacity': 0.5}
).add_to(m)

# Plot solar panel structures
folium.GeoJson(
    gdf_structures.geometry,
    name='Solar Panel Structures',
    style_function=lambda x: {'color': 'blue', 'weight': 2, 'fillColor': 'blue', 'fillOpacity': 0.3}
).add_to(m)

folium.LayerControl().add_to(m)

# Save map
map_filename = 'exeter_solar_panel_buildings_and_structures.html'
m.save(map_filename)

# Prepare summary
summary = f"There are {len(gdf_buildings)} buildings/building parts with solar panels on roofs and {len(gdf_structures)} solar panel structures in Exeter.\n"
summary += "First 5 buildings:\n" + str(gdf_buildings.head(5)) + "\n"
summary += "First 5 solar panel structures:\n" + str(gdf_structures.head(5))

return [summary, map_filename, "Folium map showing Exeter boundary, buildings/building parts with solar panels, and solar panel structures.", map_filename]
```

Tool code_executor returned result : ['There are 4468 buildings/building parts with solar panels on roofs and 1 solar panel structures in Exeter.\nFirst 5 buildings:\n osid versiondate versionavailablefromdate \n0 a578be0a-7fff-4fbb-8ace-2cf755bbb53c 2025-03-11 2025-03-12 00:00:00+00:00 \n1 c86c6a23-5ced-4dae-be17-94f33e02c66d 2025-03-11 2025-03-12 00:00:00+00:00 \n2 5cc5cbef-f15d-4110-9493-25be6ea12946 2025-03-11 2025-03-12 00:00:00+00:00 \n3 93d80cf9-5e7b-4c78-980f-c0acc04ed8a4 2025-03-11 2025-03-12 00:00:00+00:00 \n4 5b8bc540-b5d9-47bb-87bb-0d1c0a69bade 2025-03-11 2025-03-12 00:00:00+00:00 \n\n versionavailabletodate changetype geometry_area_m2 geometry_updateddate \n0 NaT New 51.920 2025-03-10 \n1 NaT New 51.910 2025-03-10 \n2 NaT New 51.910 2025-03-10 \n3 NaT New 51.912 2025-03-10 \n4 NaT New 63.966 2025-03-10 \n\n theme description description_updateddate ... nlud_code \n0 Buildings End-Of-Terrace House 2025-03-10 ... NaN \n1 Buildings Mid-Terrace House 2025-03-10 ... NaN \n2 Buildings Mid-Terrace House 2025-03-10 ... NaN \n3 Buildings End-Of-Terrace House 2025-03-10 ... NaN \n4 Buildings Detached House 2025-03-10 ... NaN \n\n nlud_orderdescription nlud_groupdescription address_classificationcode \n0 NaN NaN NaN \n1 NaN NaN NaN \n2 NaN NaN NaN \n3 NaN NaN NaN \n4 NaN NaN NaN \n\n address_primarydescription address_secondarydescription \n0 NaN NaN \n1 NaN NaN \n2 NaN NaN \n3 NaN NaN \n4 NaN NaN \n\n lowertierlocalauthority_gsscode lowertierlocalauthority_count status \n0 NaN NaN NaN \n1 NaN NaN NaN \n2 NaN NaN NaN \n3 NaN NaN NaN \n4 NaN NaN NaN \n\n status_updateddate \n0 NaT \n1 NaT \n2 NaT \n3 NaT \n4 NaT \n\n[5 rows x 122 columns]\nFirst 5 solar panel structures:\n osid toid versiondate \n0 e0f4b3bf-87e4-41f2-bb78-a4c3f3eb1a2b osgb5000005290985306 2025-07-17 \n\n versionavailablefromdate

versionavailabletodate firstdigitalcapturedate \n0 2025-07-18 00:00:00+00:00 NaT 2021-12-21 \n\n changetype geometry_area_m2 geometry_evidencedate \n0 Modified Attributes 10393.775 2024-04-20 \n\n geometry_updatedate ... nlud_orderdescription \n0 2024-10-15 ... UTILITIES AND INFRASTRUCTURE \n\n nlud_groupdescription address_classificationcode \n0 Energy production and distribution None \n\n address_primarydescription address_secondary_description \n0 None None \n\n lowertierlocalauthority_gsscode lowertierlocalauthority_count status \n0 E07000041 1.0 None \n\n status_updatedate geometry \n0 2022-09-23 MULTIPOLYGON (((-3.52322 50.70705, -3.52322 50... \n\n[1 rows x 65 columns]',]

There are 4,468 buildings/building parts with solar panels on roofs and 1 solar panel structure in Exeter.

First 5 buildings: - End-Of-Terrace House - Mid-Terrace House - Mid-Terrace House - End-Of-Terrace House - Detached House

First 5 solar panel structures: - Solar Panels On Bare Earth Or Grass

An interactive map has been generated showing the Exeter boundary, all buildings/building parts with solar panels, and the solar panel structures.

Artifact name: exeter_solar_panel_buildings_and_structures.html

Description: Folium map showing Exeter boundary, buildings/building parts with solar panels, and solar panel structures.

Artifact file: exeter_solar_panel_buildings_and_structures.html

host_agent

Here are the results for buildings in Exeter that already have solar panels on their roofs:

- There are 4,468 buildings or building parts in Exeter with solar panels on their roofs. These include various types such as end-of-terrace houses, mid-terrace houses, and detached houses.
- Additionally, there is 1 large solar panel structure in Exeter, categorized as "Solar Panels On Bare Earth Or Grass."

An interactive map has been generated showing: - The Exeter boundary - All buildings/building parts with solar panels - The solar panel structure

If you would like to view or download the map, let me know!